AllSERP: Exhaustive Per-Element Enrichment of the Versatile AdSERP Dataset

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Abstract

AdSERP [4] is a public corpus of 2,776 search trials with eye gaze, cursor, scroll, pupil and click telemetry recorded on real Google result pages, together with the page screenshots and captured HTML. Its shipped bounding boxes cover only advertisements, which receive about 16 % of main-axis clicks. AllSERP adds a typed area-of-interest (AOI) layer for everything else: screenshot-anchored boxes for organic results, People-Also-Ask, image packs, local packs and other widgets, labelled from the captured HTML. Three released flavours trade box tightness for coverage, from tight boxes to gap-filled boxes to per-card cells inside the top-ads carousel. The ad partition agrees with the shipped ad rectangles on all 38,250 comparisons, and a DOM-based harness measures box fidelity on the full corpus. Under the gap-filled flavour, 95.7 % of final clicks land in a typed main-axis AOI once cursor coordinates are converted into screenshot space. We report a per-element inventory of clicks, fixations, regressions and above-fold exposure, describe how coverage has grown across element types and within them, and release the pipeline, per-trial JSON, corpus CSVs and a browser-based replay viewer.

CCS Concepts

• **Information systems → Users and interactive retrieval**; *Web log analysis*; • **Human-centered computing** → *User studies.*

Keywords

search engine results pages, multimodal datasets, eye tracking, mouse cursor tracking, area of interest, dataset enrichment

1 Introduction

AdSERP [4] records what people looked at, moved toward and clicked on real commercial-intent Google result pages. Each of its 2,776 trials carries a full-page screenshot, the captured HTML, 150 Hz Gazepoint eye tracking, `evtrack` cursor telemetry [5], scroll and pupil signals. The pages were collected before AI Overviews reached the live SERP, so the corpus is also a fixed reference point for how those pages were used. Early work on it predicts per-slot attention from cursor trajectories [7] and develops real-time pupillometric load measures [1, 3].

The shipped geometry stops at advertisements. AdSERP's bounding boxes cover the top-ads carousel, in-column native ads and right-rail ads, and those surfaces take about 16 % of the clicks that land on the main column (about 19 % of final clicks once right-rail ads are counted). The rest land on organic results, People-Also-Ask, image packs, local packs and other widgets that have no boxes at all. Analyses that want a per-element read have had to pool ads with organics under an absolute rank, or estimate organic positions from heading counts, which assumes uniform result heights when real results run 80 to 320 px tall.

AllSERP fills that gap. It anchors boxes to the shipped screenshots, labels each box from the captured HTML, and releases the result as three flavours of one corpus CSV plus per-trial JSON. This paper describes the pipeline, what has been checked about it, what the corpus looks like per element type, how far coverage now reaches, and where it stops.

Contributions.

1. (1) A screenshot-anchored, HTML-typed AOI pipeline covering eight main-axis element types and the off-axis surfaces (§2), released in three flavours: tight boxes, gap-filled boxes, and cell-aware boxes that subdivide the top-ads carousel.
2. (2) Validation against the shipped ad rectangles (38,250 comparisons, no disagreements) and a DOM-based fidelity harness that scores the released boxes, trial by trial, on the full corpus (§3).
3. (3) A per-element inventory of click share, fixation coverage, regression rate and above-fold exposure, re-derived on the current substrate (§4).
4. (4) An account of how coverage has grown, across element types and within them, with the maturity of each layer stated so consumers can pick what to trust (§5).
5. (5) Code under MIT, derived data under CC-BY-4.0, and a replay viewer that renders the enrichment on the source screenshots (§7).

2 The Pipeline

2.1 Why the screenshot, not the DOM

The AdSERP team extracted their ad boxes from the live DOM at collection time, which is the cleanest source while the page is live [4]. The route does not survive reuse. Re-rendering the saved 2022–2023 HTML in a 2026 browser drifts by about 13 px at the median and about 45 px at the page bottom relative to the original screenshots, which were captured on Chrome 110 on Windows. Missing assets, browser versions, fonts and Google's own layout experiments all contribute. The fixation and cursor streams are coordinated against the screenshots, so AllSERP measures geometry on the screenshots and uses the HTML only for labels.

2.2 Four phases

Phase A finds card spans by per-row standard-deviation projection on the main column. Shipped ad rectangles take precedence where they overlap, and a composite-height trigger subdivides image packs, People-Also-Ask and top-stories blocks. **Phase B** walks the saved HTML and assigns each card a type through an eight-tier priority chain, from heading text through structural class signatures and `data-attrid` values to fallback patterns. **Phase C** binds labels to geometry by document order, the one axis the HTML and the screenshot share, then verifies the pairing geometrically (§5 describes what that verification changed). **Phase D** extends each pair

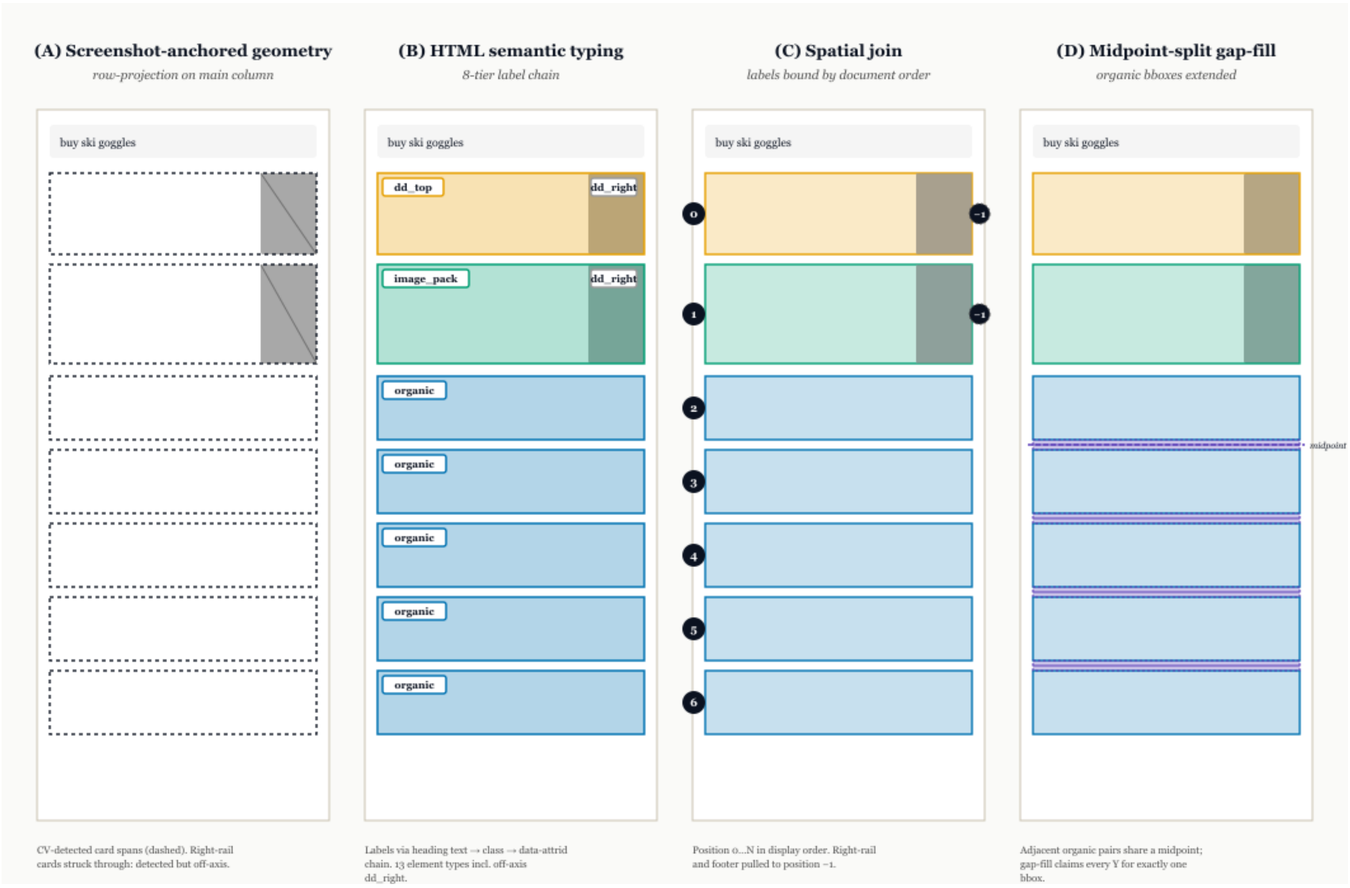


Figure 1: The four phases on a synthetic SERP. (A) Row projection on the main column yields card spans. Right-rail cards are detected but routed off-axis. (B) Each card receives a label from the captured HTML. (C) Labels bind to geometry in document order. Main-axis cards get positions $0 \ldots N$; right-rail and footer cards get -1. (D) Adjacent organic boxes extend to their shared midpoint so every y in the column belongs to one box. Ads and widgets are unchanged.

of adjacent organic boxes to their shared midpoint, clamped so an organic never crosses an ad or a widget. The midpoint is a heuristic for the boundary between two results, not DOM truth. Misattribution is bounded by the inter-result gap, typically 5 to 60 px, and both the tight and the gap-filled boxes are released so a consumer can pick.

The main-axis label set is `organic`, `dd_top` (top-ads carousel), `native_ad`, `paa`, `image_pack`, `top_places` (local pack), `other_widget` and `unknown_widget`. Right-rail ads (`dd_right`), right-rail knowledge panels, related searches and page chrome are labelled but carry position -1. Widget blocks are first-class slots in display order: a local pack occupies a main-axis position, and 353 of the 2,776 trials (12.7 %) contain one. The `organic_rank` column numbers organics only, so a widget slot never shifts an organic's rank, and the producer measures that rather than asserting it (zero misshifted organics on the shipped maps). Twelve of those 353 trials are on the exclusion list and one has no fixations, which is why Table 2 carries 340 local-pack AOIs.

2.3 Three flavours

Table 1 lists the released flavours. The cell-aware flavour applies on the x axis what Phase D applies on y: a horizontal carousel's cells are widened across their inter-card gaps so a click in a card margin attributes to the nearest cell rather than being lost.

2.4 Click attribution and coordinate spaces

A click is attributed to a main-axis AOI when both its x and y fall inside the box, with a ± 5 px x and ± 10 px y tolerance for link padding. Requiring x matters: y-only attribution routes right-rail ads, page chrome and far-off-target clicks into whichever main-axis box shares their y. A trial-level helper, `is_main_axis_click`, reports whether a trial's final click lands in a main-axis AOI under this rule, and producers that compute click-outcome features drop the trials it flags.

The two instruments do not write into the same space. AdSERP records three widths per trial (screen 1280, document 1403, window 1422 px). Gazepoint is hardware and reports screenshot space, so fixations need no conversion. `evtrack` is JavaScript and reports

Table 1: Released flavours of the corpus CSV. Each is one file with the same schema.

Flavour	What it is, and when to use it
`typed`	Phases A to C. Tight boxes on visible text. Use when a fixation must be attributed conservatively.
`typed_gapfill`	Phases A to D. Organic boxes extended to their midpoints. **Recommended default** and the basis of §4.
`typed_gapfill_cellsplit`	Superset of the gap-fill flavour. Adds `role = cell` rows that subdivide the top-ads carousel into its cards, sparse organic sub-cells, and the off-axis right-rail block as a covariate. Filtering `role = parent` and `main_axis` recovers `typed_gapfill` exactly.
`organic_hybrid`	Three labels only (`organic`, `native_ad`, `dd_top`) with all-main-axis position numbering. Built from Phase A geometry and the shipped ad rectangles, so it does not depend on the HTML typing layer. Use for continuous position-rank analyses where the topmost SERP slot is the unit.

document space, so cursor and click samples do. The released filter, as shipped, compares unconverted `evtrack` coordinates against screenshot-space boxes. On that path, 237 trials are flagged: 67 final clicks inside a shipped right-rail ad rectangle, 169 on page chrome, search tools or far off any box, and 1 with no usable click, for a corpus-level attribution rate of 91.5 % (2,539 of 2,776). Converting each final click through its own trial's document-to-screenshot ratios before the containment test recovers 118 of the flagged trials and loses none, for **95.7 %** (2,657 of 2,776). The chrome and off-target category falls from 169 to 15, while right-rail ad clicks rise from 67 to 103: 36 of the supposed off-target clicks were right-rail clicks whose unscaled x sat inside the main column's apparent range. That no trial attributes under the shipped path and fails under the converted one is why we read the conversion as a correction rather than a different answer.

The same conversion reframes why the gap-fill flavour was built. Under tight boxes and document-space clicks, 23.9 % of approached-and-clicked records had their click land outside the box that was approached. In screenshot space that contamination is 3.9 % (94 of 2,406 records). The 510 records once classed as bbox-edge near misses drop to 5. They were right-rail clicks whose unscaled x fell inside the main column's apparent range. The midpoint split still recovers clicks that land in the padding between results, but most of the problem it was built to fix was a coordinate bug.

All click numbers in this paper are reported in screenshot space unless a sentence says otherwise. The shipped `is_main_axis_click` helper still runs in document space, because re-basing it moves every click-outcome population downstream of it. That re-basing is a substrate revision, and the per-trial ratios needed to apply it are in the released metadata.

3 Validation

3.1 Agreement with the shipped ad rectangles

The shipped ad rectangles were extracted by the AdSERP authors against the same screenshots the gaze and cursor streams were recorded against, so they are a usable gold standard for the ad-versus-non-ad partition. Two checks run against them. None of the 26,590 Phase A organic boxes overlaps a shipped ad rectangle. Every one of the 11,660 shipped ad rectangles (9,217 native ads, 1,582 top-ads carousels, 861 right-rail ads) emerges as a typed entry of the matching label, and every typed ad entry matches a shipped rectangle, with mean IoU 1.000. Together these are 38,250 classifications with no disagreements. The check is bounding box against bounding box and admits no cursor coordinate, so it is indifferent to the coordinate question of §2.4, and it holds unchanged on the current substrate. It is an internal-consistency check, not independent annotation: Phase C inherits ad identity by spatial overlap and is then compared to the same rectangles. The non-ad partition (organic versus widget types) has no external label set. It is checked against the HTML structure the typing chain consumes, against the DOM harness below, and by visual inspection in the replay viewer. Independent annotation of a held-out subset remains future work.

3.2 DOM fidelity harness

Box quality is measured rather than asserted. A harness (`aoi_fidelity.py`) renders each saved SERP, resolves each released box to the DOM element it claims, and scores three per-trial checks over the full corpus. The render is subject to the drift of §2, so the IoU check is bounded by that drift and reads as a floor:

- the recorded final click falls inside the element its xpath names on 87.7 % of trials (2,433 of 2,775);
- a released box matches its DOM element at IoU $\geq$ 0.5 on 90.8 % of trials (2,520 of 2,776), median IoU 0.879;
- the visible top-carousel card count agrees with the cell export on 29.5 % of carousels (466 of 1,582), the weakest layer, discussed in §5.

Two cautions travel with these rates. A 120-trial retained sample scores about 2.5 points higher on the first two checks, so sample and corpus figures must not be quoted interchangeably. And the box rate was 93.4 % before the card-collision fix of §5 changed 454 maps. No full-corpus pre-fix output survives to diff against, so we disclose the gap rather than attribute it.

3.3 Gaze–cursor registration

The resource also depends on gaze and cursor being registered to each other. For each trial with a final click and at least one fixation in the 1,500 ms before it (n = 2,752), we take the minimum distance from any such fixation to the click. The median is 128.8 px (interquartile range 80.4 to 206.5 px, 95th percentile 446.2 px). In 17.8 % of trials (489) no fixation in the lead window came within 250 px, about 3° of visual angle at the AdSERP viewing distance and past the tracker's nominal calibration tolerance. The distance at the instant of the click is not the right probe, because gaze leads

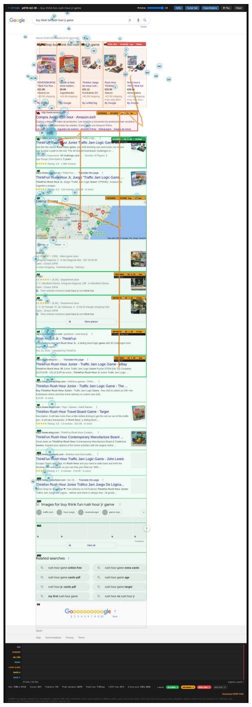


Figure 2: Replay viewer rendering of trial p010-b2-t6 (§7). Coloured rectangles: typed AOIs with their consideration-set label. Numbered circles: gaze fixations sized by dwell. Orange polyline: cursor trajectory. Timeline tracks: AOI under cursor, cursor position, cursor speed, pupil diameter, pupil LF/HF ratio on a log scale, gaze x and y. Each track carries its range in units.

the cursor by several hundred milliseconds on SERPs. That concurrent median is 506.8 px and is reported only to separate the two questions.

4 Element-Type Inventory

Table 2 reports the inventory under the gap-fill flavour on the current substrate: 2,764 trials after alignment exclusions (§7), of which 2,763 processed, yielding 36,396 typed main-axis AOIs. Four denominators appear in this paper. Counts against the shipped ad rectangles and the trial-level click filter use all 2,776 AdSERP trials, the fidelity harness uses 2,775 or 2,776 depending on the check, and the typed maps use the 2,764 retained trials, 2,763 once one trial without fixations is dropped. Knowledge panels do not appear because on these pages they render in the right rail and are now routed off-axis (§5). Column denominators differ and are given in the notes.

Column notes. *Fixated %* is over the element type's AOIs. *Clicks* counts every click event landing in an AOI of that type, and *click %* is that count over the 2,749 clicks attributed to any main-axis AOI in screenshot space (2,627 in document space). The conversion is not a uniform gain: it moves 26 of the 29 clicks that document space placed on a People-Also-Ask box to the organic result directly above it, the one-card shift that a 10 % y overshoot produces. The click xpaths that `evtrack` recorded name organic-result headings for those clicks, so the converted attribution is the right one. The corpus-level rate in §2.4 counts trials whose *final* click lands main-axis under the trial filter, a different denominator. *Regressive %* is the share of fixated AOIs that were returned to by a regressive fixation, over the fixated subset, and so is not directly comparable across types whose fixation rates differ widely. *Above-fold %* is the share of trials in which at least one AOI of that type sits in the initial viewport.

Reading the table. The top-ads carousel is fixated on nearly every trial it appears in but takes about a tenth of clicks. Native ads show the same shape at lower amplitude, 36 % fixated and 6 % of clicks. The dissociation between attention and clicks that the original AdSERP analysis reported at the ad-versus-organic level survives at element granularity. Part of the carousel's fixation rate is geometry: it is large and above the fold on 57 % of trials, so fixation coverage should be read alongside AOI area rather than as attentional pull. Organic results are above the fold on 94 % of trials and take 80 % of clicks. Local packs, now typed on 340 AOIs rather than the 84 of earlier releases, are fixated at 91 % and returned to regressively at 76 %, the highest regressive rate after the carousel. People-Also-Ask is the other extreme: fixated on 44 % of its AOIs but clicked twice in the whole corpus once clicks are placed in the maps' coordinate space. On this task it is read, not used.

Inside the top-ads carousel. The cell-aware flavour resolves the carousel into its cards. In the released cell snapshot, 1,550 of 1,582 carousels subdivide, into a median of four cells (range 2 to 6, modal 4 on 61.9 %). Of the 237 final clicks landing inside a carousel, the leftmost cell draws 31.7 % on the modal four-cell layout ($n = 142$) against a 25 % uniform baseline, with the other cells near-even at 21 to 24 % (Fig. 4). Pooling across carousel sizes and indexing by cell rank, click-through falls from 4.3 % at the leftmost cell to 2.0 % at the fifth (231 ranked clicks), and mean fixation count and dwell fall with it (7.4 to 3.8 fixations, 1.6 to 0.9 s). Each decline is directional, not monotone: the first and second cells are not reliably ordered for clicks under any denominator we tried, and the fourth cell sits above the third on fixations and dwell, so we report endpoints and no rank coefficient. A coefficient over six cell means, the sixth resting on five carousels, would say little.[1] These numbers come from a frozen cell snapshot whose card boundaries are only partly certified, as §5 explains, and they will be re-derived when the DOM-derived cell source replaces it. The block-level carousel box cannot express either distribution. The subdivision is what makes within-surface position legible at all.

5 Coverage: How Far It Reaches, and Where It Stops

A resource like this is built in layers, and each layer has a different amount of evidence behind it. Table 3 states the layers and their

[1] Version 3 of this paper reported $\rho = -1.0$ for the click decline and $\rho = -0.94$ for fixations and dwell, all computed over six cell means. The perfect monotone does not survive the 2026-08-30 substrate audit, and the fixation and dwell coefficients rest on the same six points, so all three are withdrawn. The same audit's first cell-agreement figure, 29.1 %, is also withdrawn: its counter mixed cell types and omitted zero and missing comparisons. The corrected full-corpus figure is the 29.5 % in §3.

Table 2: Per-element-type behavioural inventory under the gap-fill flavour. Element types ordered by AOI count. Click columns use screenshot-space coordinates.

element type	AOIs	fixated %	clicks	click %	regressive %	above-fold %
organic	22,269	55.2	2,232	81.2	56.8	93.8
native_ad	9,180	36.4	156	5.7	46.8	37.7
dd_top	1,575	**99.7**	280	10.2	83.6	57.0
image_pack	1,566	50.3	40	1.5	61.7	19.2
paa	765	43.5	2	0.1	50.8	11.0
unknown_widget	673	18.1	14	0.5	32.0	0.5
top_places	340	91.5	24	0.9	76.2	11.8
other_widget	28	39.3	1	0.0	27.3	0.3

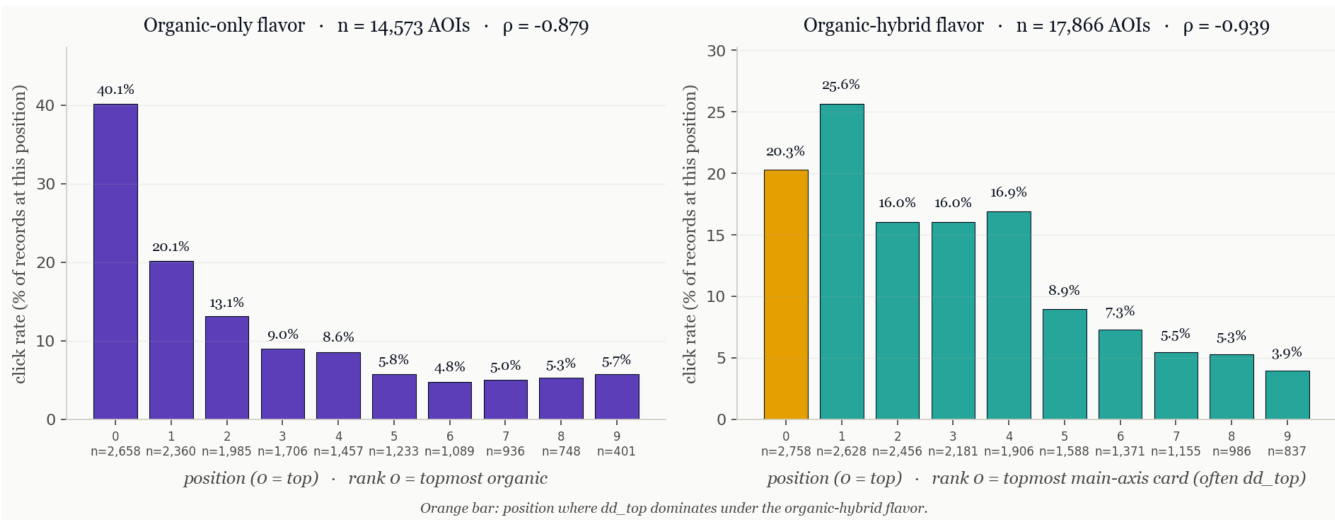
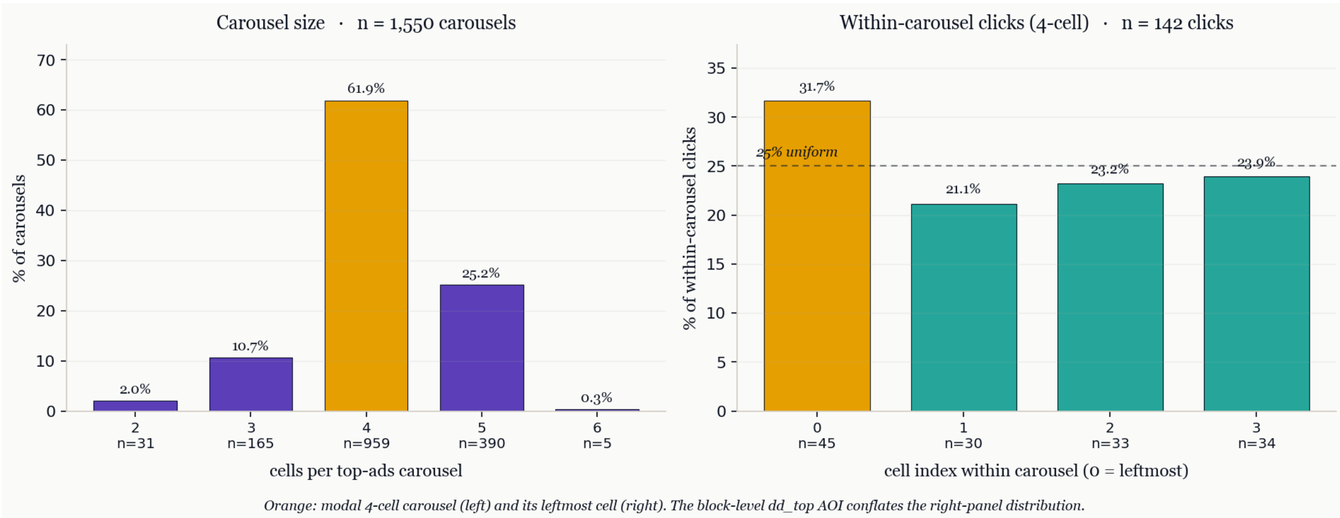


Figure 3: Click rate by SERP position under two position conventions, on the current feature files (positions 0 to 9). Per-bar n is the number of AOI records at that position and click rate is computed within that bar only. Left, organic-only positions (14,573 records). Position 0 is the topmost organic, with a 40.1 % click rate, declining to 5 to 6 % from position 5 on (Spearman $\rho = -0.879$ over the ten position rates). Right, all-main-axis positions (17,866 records). Position 0 is the topmost card, most often a top-ads carousel (orange, 20.3 %), and the click peak moves to position 1 where the top organic now sits (25.6 %, $\rho = -0.939$). Both panels use the `organic_hybrid` and organic flavours, which are built from Phase A geometry and the shipped ad rectangles and do not depend on the typing layer. Pick the convention that matches the question: what the top organic earns, or what the topmost slot earns.

status. This section records how the coverage got here, because the history explains the caveats.

Across element types. Coverage grew in steps. AdSERP shipped ads. The first AllSERP layer added organic boxes from the screenshots. The second added HTML typing for the widgets that had until then been pooled with organics or dropped. The third added the midpoint gap-fill, after an audit showed that tight boxes lost clicks landing in the padding between results. The fourth, released as AllSERP v1.1.0, replaced index-based label-to-box pairing with geometric verification. Geometric verification found two errors in the earlier maps. Local packs render outside the main results container, so the typing pass never emitted a card for them while the box extractor did, and the box took the next card's label, pushing every card below it one slot down. Right-rail knowledge panels, which render at $x \approx 804$ on a 1,280 px page, had been given a main-column slot they never visually occupied. Correcting both changed the main-axis label sequence on 42 % of trials, moved local packs from 84 typed boxes to 340, removed 746 knowledge-panel rows from the main axis, and excluded 12 trials whose rank lattice cannot be verified geometrically. A card-collision fix on 2026-08-30 then removed 454 boxes that duplicated another card's geometry. Aggregate rank medians were robust to all of this (per-rank LF/HF medians moved by under 0.6 at every rank). Per-trial labels were not, which is why every number in this version was re-derived.

Within element types. The top-ads carousel is the first surface subdivided below the card. The released cells come from a snapshot frozen in May 2026, aligned to the parent carousel box but not verified card by card. The DOM harness's count check makes



Orange: modal 4-cell carousel (left) and its leftmost cell (right). The block-level dd_top AOI conflates the right-panel distribution.

Figure 4: Anatomy of the top-ads cell split, from the released cell snapshot. Left: carousels hold 2 to 6 cards, modal 4 (61.9 % of 1,550). Right: of the 237 final clicks landing inside a carousel, the distribution on the modal four-cell layout (n = 142) favours the leftmost cell (31.7 % against a 25 % uniform baseline) but spreads across all cells.

Table 3: Coverage layers in the release, from the surfaces AdSERP shipped to the ones still being built. Counts are on the current substrate unless a row says snapshot.

Layer	Status	Coverage	What certifies it
Across element types (one box per card)			
Ads: carousel, native, right rail	shipped by AdSERP	11,660 boxes	matched one-to-one by AllSERP, 0 disagreements
Organic results	released	22,269 main-axis boxes	DOM harness, 90.8 % at IoU $\geq$ 0.5. The midpoint gap-fill is a heuristic
Widgets: PAA, image packs, local packs, other	released	3,372 main-axis boxes	HTML structure plus DOM harness. No external label set
Right-rail knowledge panels	off-axis	position −1	screenshot-verified as right rail, not a main-axis slot
Within element types (cells inside a card)			
Top-ads carousel cards	released, frozen snapshot	6,373 cells in 1,550 trials	internal alignment to the parent box only. Visible card count agrees on 29.5 %
Top-ads carousel cards, DOM-derived	candidate, not yet released	7,265 cards, 1,570 of 1,575 subdivisions admitted	screenshot registration of card borders. The 5 rejected cases stay in the denominator
Organic sub-cells	preliminary	174 cells in 75 trials	45.8 % of candidates align to an organic box. Only the aligned subset ships
Right-rail cells	covariate	184 cells in 92 trials	off-axis, shipped so main-axis models can condition on right-rail exposure
Sitelinks and other sub-card structure	not yet	—	—
AI Overview cards	not applicable	—	corpus predates the rollout; a label would be added in Phase B

the limitation concrete: visible card counts agree with the snapshot on 29.5 % of carousels, with 961 short and 155 over. A replacement source derives cells from the DOM and registers each card border against the original screenshot. On the full corpus it admits 1,570 of 1,575 eligible carousels with matching independent counts, 7,265 cards in all, and the 5 rejected cases stay in the denominator. It is not yet the released source, because adopting it requires a versioned cell-source contract so that the per-cell numbers in §4 are re-derived rather than silently replaced. Organic sub-cells and right-rail cells are shipped at the maturity Table 3 states.

Paging is rare enough to define exposure by the viewport. A typical carousel is a horizontal scroller several times wider than its viewport (one audited page: 3,796 px of cards clipped to about 652 px, 27 cells in the DOM, about 5 visible). In the 2026-08-30 audit, which counted every click event landing in a cell (272 events, a broader count than the 237 final clicks above), no click landed on a card that was not already on screen, including in the 158 trials where the carousel had been scrolled. For the cell layer we therefore treat the visible cards as the cards a participant was exposed to. Snapshot visibility alone does not record paging history, and the audit's selector scope is being re-verified before this definition is built into the DOM-derived source.

6 Limitations

Task design. AdSERP elicits one click per query within a one-minute window (extended to two after a prompt) on a single query phrase per trial. The corpus does not sample reformulation, pagination, abandonment, multi-query sessions, or decision arcs longer than that envelope. The statistics in §4 describe constrained-choice SERP evaluation, not free search.

A pre-AI-Overviews snapshot. Data were collected between 2022-12-16 and 2023-03-13 (from per-trial timestamps), before Google's AI Overviews rollout (limited test May 2023, broad deployment May 2024) and eight days before Bard opened to the public on 2023-03-21. The corpus is therefore a clean pre-rollout baseline, and its value as a reference grows as the live SERP diverges from it. Replications on later SERPs need an `ai_overview` label, and the AI answer card has different click and fixation dynamics. The pipeline shape is era-agnostic. Only Phase B's label set needs to grow.

Heuristics and missing labels. The midpoint split and the cell widening are heuristics with bounded error, and both tight and extended boxes ship. The non-ad partition has no independent annotation. The right rail's organic-style results are not recovered, and about 1 % of trials have a right-rail click with no shipped rectangle, a property of the original release.

7 Release and Data Statement

7.1 Artifacts

- **Code.** `attentional-foraging` (https://github.com/andyed/attentional-foraging) and `approach-retreat` (https://github.com/andyed/approach-retreat), MIT licence. Single entry point: `scripts/build_aois.py--flavor<name>`.
- **Derived data (CC-BY-4.0).** The four corpus CSVs of Table 1, matching per-trial JSON (about 19 MB across the corpus), and a per-(trial, position) content-feature file with lexical, query-overlap and semantic-cosine features. Snippet lexical diversity is correlated with rank, so control for rank before claiming a per-position content effect.
- **Replay viewer.** https://andyed.github.io/approach-retreat/replay/ renders typed AOIs, fixations, cursor and the pupil, LF/HF and gaze timelines on the source screenshots for 148 curated trials (Fig. 2). Two of the 148 are on the alignment-exclusion list and are badged as such.
- **Underlying corpus.** AllSERP requires the AdSERP Zenodo volume (https://zenodo.org/records/15236546), released CC-BY-4.0. AllSERP does not redistribute the corpus. The replay viewer renders screenshots for its 148 curated trials, and the carousel audit evidence carries card crops and captured card text for the trials it inspected, both under the corpus licence with attribution.

The corpus CSV is the join target. Existing AdSERP signal files (cursor approach features, pupil power ratios, saccade orientation) merge on `trial_id` and gain a per-AOI `etype` column without re-collecting data. The repository README carries a short Polars example.

7.2 Substrate identity

The maps this paper describes are AllSERP enrichment **v1.1.0**, pinned by content rather than date: 2,764 analysable trials, an alignment-exclusion list of 12 trials with its rule (`mean_residual > 30` or `mean_h_mismatch > 60 and mean_residual > 8`), and the typed-map content hash `2cb789eb8febd234`. The exclusion list is quarantine, not relabelling, and its membership changed when it went from 14 to 12 entries, so a consumer should re-read it rather than diff the count. Every export summary embeds the list. A quick tell for a stale export: earlier maps contain about 746 main-column knowledge-panel rows and 84 local packs, and v1.1.0 contains 0 and 340.

7.3 Changes since version 3

- Every number re-derived on v1.1.0. Table 2 loses its knowledge-panel row and gains 256 local-pack AOIs. Click columns now use screenshot-space coordinates.
- The within-carousel click decline is restated as directional. The $\rho = -1.0$ of v3 is withdrawn (§4).
- The coordinate-space finding of §2.4 is new, as are the DOM fidelity harness rates and the coverage account of §5.

7.4 Reproducing the numbers

All figures are produced by scripts in the `attentional-foraging` repository. Table 2: `allserp_descriptives.py--flavortyped_gapfill--spacescreenshot`. Trial filter and conversion: `audit_trial_filter_space.py`, with `audit_dd_right.py` and `audit_cascade_contamination.py` (each with `--space`) for the right-rail and contamination counts. Ad-rectangle agreement: `validate_typed_ads_vs_shipped.py`. DOM harness: `aoi_fidelity.py--all`, whose full-corpus output ships as `aoi_fidelity_full_2026-09-07.json`. Gaze-cursor registration: `audit_gaze_cursor_coverage.py`. Cell composition and rank: `cellsplit_click_composition.py` and `compute_nb23_cellsplit_rank.py`, with the count audit's evidence under `docs/evidence/carousel-2026-09-04/`. Each script prints the coordinate space it ran in.

7.5 Citation and provenance

AllSERP ships only AOI labels, geometry and the pipeline. The boxes are pixel-anchored to AdSERP screenshots and indexed by AdSERP trial IDs, so any analysis that joins them to gaze, cursor, scroll or pupil streams needs the AdSERP corpus, and both citations apply [4]. Applying the pipeline to a different SERP corpus is the one use for which the AllSERP citation alone suffices. AdSERP collection followed the original team's approved IRB. AllSERP processes only de-identified telemetry and screenshots already released under CC-BY-4.0.

8 Conclusion

AllSERP gives every card on an AdSERP page a box and a label, checks the ad partition against the shipped rectangles, measures the rest against the DOM, and says which layers are certified and which are not. With element type and geometry joined to the multimodal telemetry, per-element questions become tractable on the existing corpus: cursor approach and retreat per AOI rather than per rank, pupillometric load conditioned on element type [1, 3], and content features paired with behavioural outcomes on the surface where they occurred, with information-foraging measures [2, 6] as one natural lens. The inventory here is the descriptive baseline for that work on a pre-AI-Overviews snapshot of commercial search, and the pipeline shape carries to later SERPs with a larger label set.

9 AI Use Disclosure

This paper was drafted and revised with Anthropic's Claude (Opus 4.7 for the initial draft, Opus 4.8 for the cell-split revision, Fable 5.1 for this restructure). Automated passes checked citations against the bibliography and literature notes, flagged ungrounded comparatives and causal language, and cross-checked numbers between sections, tables and captions. All quantitative claims trace to scripts in the released repository. The author verified every claim, citation and methodological decision and takes full responsibility for the content.

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